Problem Statement and Objectives

- Design an AI system that **infers cognitive states** from EEG signals recorded during different sensory stimulation conditions (auditory, gustatory, olfactory).
- Objectives:
  - **Problem Formulation**
  - **Machine Learning**
  - **Probabilistic Reasoning**
  - **Comparative Analysis**
  - **Ethical Reflection**

Experiment Design and Results

EEG data was extracted from multiple channels and frequency bands, including delta, theta, alpha, beta, and gamma. Overall, the CNN was the strongest model as it had the highest accuracy in both experiments, with a 0.672 and 0.837 in experiment 1 and 2. It also had the 2nd highest AUC-ROC overall, with 0.841 and 0.888 in experiment 1 and 2. The decision tree was the second strongest, as it had a 0.606 and 0.468 accuracy in both experiments, and a 0.872 and 0.737 for AUC-ROC. The probabilistic models were helpful for modeling uncertainty and hidden state transitions.

Methods

Over the two experimental datasets we used to complete this project, one being music stimulation in various states, and the other being physical stimulation using things like coffee and perfume. To analyze the data we used three different types of AI, the decision tree classifier, the convolutional neural network, then finally probabilistic reasoning to model certain forms of uncertainty in the EEG data that was collected. In addition, all our experiments were done locally on one machine with the same environment so all outside variables were eliminated to the highest degree we could.

Conclusion and Future Work

This project demonstrated that EEG signals recorded using a consumer-grade headband contain sufficient information to classify sensory stimulation, specifically auditory, gustatory, and olfactory stimuli. Among the evaluated models, the CNN achieved the highest performance, followed by the decision tree and Naive Bayes classifiers. However, uncertainty remains in these results due to inter-individual variability and other uncontrollable factors that may have influenced the recordings.

Future work should focus on expanding the dataset size and incorporating additional physiological signals to improve classification accuracy, robustness, and generalizability.